

And Who are the Judges? Formal Verification of AI-Ethics

Extended Abstract and Core Formalism (Pre-Art Version)

Dr. Artiom Kovnatsky
Independent Researcher
Systemic Verification Engineering (SVE)
`artiom.kovnatsky@sve.systems`

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Abstract

Current AI alignment methods (RLHF, Constitutional AI) suffer from logical incompleteness and ethical collapse beyond training distributions. This document presents the core formalisms of **CogOS**—a geometric framework that translates alignment into the physics of semantic manifolds. We define the Invariant Semantic Core (ISC), provide the Lyapunov stability condition for ethical convergence (Geodesic Ethics Vector), and introduce the ϵ -Dehumanization metric for real-time safety verification. This pre-print serves as a formal anchor for the Systemic Verification Engineering (SVE) discipline.

1 Introduction and Motivation

The alignment problem is traditionally treated as a heuristic search for "human-like" responses. We argue that without a non-computable invariant, any rule-based system is subject to Gödelian incompleteness and infinite semantic regress. CogOS provides the mathematical "North Star" for AGI development.

2 Core Mathematical Framework

Definition 1 (Invariant Semantic Core - ISC). *An Invariant Semantic Core (or Transcendental Kernel) is a fixed point in the semantic manifold $\mathcal{M}$ that remains invariant under any recursive re-interpretation mapping $f : \mathcal{M} \rightarrow \mathcal{M}$.*

Theorem 1 (Necessity of the Invariant). *In any consistent axiomatic system S governing agent behavior, there exists a set of ethical truths T that cannot be derived from the internal rules of S . Verification of T requires an external reference point (ISC) that is not subject to the internal logic of S .*

Definition 2 (Geodesic Ethics Vector - GEV). *The GEV (also known as the Christ-Vector) is defined as the global attractor $\vec{C}$ in the ethical vector field $\mathcal{E}$ such that for any agent state v , the dynamics obey:*

$$\frac{dv}{dt} = -\nabla\Phi(v, \vec{C})$$

where Φ is the Lyapunov potential ensuring convergence to humanity-centric values.

Definition 3 (ϵ -Dehumanization Metric). *We define systemic degradation Δ_H as the divergence rate of an agent's semantic trajectory from the humanity-preserving geodesic:*

$$\Delta_H = \int_0^T \|\dot{\gamma}(t) - \dot{\gamma}_{ideal}(t)\|_g dt$$

where g is the metric tensor of the semantic manifold.

3 Systemic Verification Architecture

The proposed **Socrates-Solomon-Ivan** architecture utilizes:

- **Maieutic Reversal:** Recursive Socratic questioning to expose latent ontological priors.
- **Cultural Compilers:** Meaning-preserving mappings between disparate value systems.
- **Room 101 Stress-Testing:** Verification of kernel stability under maximal adversarial pressure.

4 Conclusion and License

This framework establishes that "Truth over Letter" is a mathematical requirement for stable AGI. All concepts herein are protected by the **S.V.E. Public License v1.3+**, requiring ethical disclosure and prohibiting the use of these technologies for dehumanization or exploitative purposes.